

Functional coupling between MC4R and Kir7.1 contributes to clozapine-induced weight gain in female mice

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Most antipsychotic drugs (APDs) cause hyperphagia and weight gain, yet the neural mechanisms underlying these metabolic side effects remain elusive, in part due to difficulties in modeling them in rodents. Here, we establish a mouse model that recapitulates clozapine-induced metabolic syndrome, enabling mechanistic investigation of this widely prescribed APD. We show that clozapine promotes obesity in female mice by driving hyperphagia, which requires functional coupling between the melanocortin 4 receptor (MC4R) and the Kir7.1 potassium channel. Within the broader context of clozapine's polypharmacology, this signaling axis emerges as a critical downstream convergence node for APD-induced metabolic dysfunction. Mechanistically, clozapine inhibits MC4R-expressing neurons in the paraventricular nucleus of the hypothalamus by enhancing MC4R-Kir7.1 coupling, thereby increasing inward potassium currents. Notably, clozapine produces this inhibition without binding the MC4R orthosteric site and without engaging canonical G α s signaling. Genetic deletion of Kir7.1 in MC4R neurons or pharmacological inhibition of Kir7.1 reverses clozapine-induced weight gain while preserving its behavioral efficacy in established antipsychotic assays. Together, these findings reveal a G-protein-independent mechanism by which clozapine disrupts energy balance and identify MC4R-Kir7.1 coupling as a therapeutically tractable pathway for mitigating APD-associated metabolic dysfunction.

Antipsychotic drugs (APDs) demonstrate broad efficacy in treating diverse neuropsychiatric conditions¹. However, most APDs induce metabolic syndrome characterized by hyperphagia, weight gain, and insulin resistance². Despite interventions such as lifestyle modification, nutritional counseling, and weight-loss pharmacotherapy^{3–5}, these metabolic side effects remain difficult to manage and are a leading cause of treatment discontinuation.

APDs act on multiple neurotransmitter receptors throughout the brain^{6–8} and this polypharmacology is central to both their therapeutic

benefits and metabolic liabilities. While their therapeutic efficacy primarily stems from blockade of dopamine D2 and serotonin 2a receptors^{9,10}, the metabolic side effects of clozapine, olanzapine, and risperidone are thought to arise from convergent action across a broader receptor set that includes high-affinity antagonism at hypothalamic histamine H1, serotonin 5-HT2C, dopamine D2/D3, and muscarinic M3 receptors^{11–15}, alongside effects on MC4R-dependent melanocortin signaling¹⁶. These actions are thought to perturb hypothalamic energy-balance circuits and promote hyperphagia and

metabolic dysfunction^{17–19}. Disentangling the contribution of any individual pathway within this polypharmacological landscape has remained a central challenge.

A major obstacle to mechanistic studies has been the inability to reliably reproduce these metabolic effects in rodent models. For example, clozapine, one of the most effective and commonly prescribed APDs, causes pronounced weight gain in both adolescents and adults²⁰, yet chronic administration of clozapine to rodents via intraperitoneal injection or drinking water produces only modest effects on food intake and body weight^{21,22}. In this study, we establish a mouse model that faithfully reproduces clozapine-induced weight gain and identify the interaction between the melanocortin-4 receptor (MC4R) and the Kir7.1 potassium channel as a key mediator of this effect. While other receptor pathways engaged by clozapine likely contribute to its overall metabolic impact, our findings establish MC4R–Kir7.1 coupling as a required convergence node within this broader polypharmacology.

Results

A mouse model of clozapine-induced metabolic syndrome

To model clozapine-induced metabolic dysfunction, we formulated a clozapine diet by compounding clozapine (50 milligrams per kilogram) into a control diet (Research Diets, Inc. D09092903). When scaled by body surface area, this dietary dose in mice is equivalent to approximately 4 mg/kg in humans (~300 mg/day for a 70 kg individual)²³, which falls within the lower end of the clinical dosing range (typically 300–600 mg/day). Consistent with this, plasma clozapine concentrations after four weeks averaged $1.12 \pm 0.35 \mu\text{g/mL}$, within the therapeutic plasma range reported in patients²⁴. The clozapine and control diets were identical in macronutrient composition and energy density (Supplementary Table 1). Although plasma clozapine levels were markedly elevated in treated mice, they showed no preference between the two diets (Fig. 1a and Supplementary Fig. 1a), indicating that clozapine supplementation did not alter palatability. Previous studies have shown that the metabolic effects of APDs are more pronounced in female than in male C57BL/6 mice^{16,25,26} (Supplementary Fig. 1b). We therefore focused on female C57BL/6 mice to assess both the chronic and acute effects of clozapine on energy metabolism.

Female mice fed the clozapine diet gained significantly more weight than controls over a 12-week period (Fig. 1b). Nuclear magnetic resonance (NMR) analyses revealed that the excessive weight gain was largely due to increased fat mass (Fig. 1c), evident in both inguinal and gonadal white adipose tissues (iWAT and gWAT respectively), but not in the brown adipose tissue (BAT, Fig. 1d). Consistent with these findings, hematoxylin and eosin (H&E) staining showed increased lipid accumulation in iWAT and liver of clozapine-fed mice (Fig. 1e, f). These mice also exhibited higher plasma levels of leptin, non-esterified fatty acids (NEFA, Fig. 1g, h) and impaired glucose clearance during a glucose tolerance test (GTT, Fig. 1i). Collectively, these findings demonstrate that dietary clozapine induces robust metabolic dysfunction in mice, closely recapitulating key features of clozapine-induced metabolic syndrome observed in humans.

Clozapine-induced hyperphagia drives weight gain

To elucidate the mechanism underlying clozapine-induced weight gain, we monitored food intake and energy expenditure in individual mice before and after clozapine treatment using a fully automated metabolic chamber system (TSE PhenoMaster). Mice were maintained on the control diet during acclimation and the first 2.5 days of the recording, followed by a switch to the clozapine diet for seven days. Remarkably, mice quickly developed hyperphagia following the dietary switch, consuming substantially more food during the dark phase of the day (Fig. 2a). Clozapine-fed mice also exhibited a modest increase in energy expenditure (Fig. 2b), whereas physical activity and

respiratory exchange ratio (RER), an indicator of substrate utilization, remained unchanged (Fig. 2c, d). Despite the slight rise in energy expenditure ($1.18 \pm 0.23 \text{ kcal/day}$), it was insufficient to counterbalance the marked increase in energy intake ($11.75 \pm 3.17 \text{ kcal/day}$), resulting in a positive energy balance and progressive weight gain.

Clozapine does not affect canonical MC4R signaling

Genome-wide association and candidate gene studies in humans have implicated the melanocortin 4 receptor (MC4R)²⁷, among others^{28–33}, in APD-induced weight gain. MC4R encodes a receptor for two hypothalamic neuropeptides with opposing effects on food intake: α -melanocyte-stimulating hormone (α -MSH) activates MC4Rs to suppress food intake, whereas agouti-related peptide (AgRP) binds the same receptor to stimulate consumption³⁴. However, whether APDs such as clozapine can directly compete for the orthosteric binding site of MC4R had not been previously reported.

In a radioligand binding assay, PF-00446687³⁵, a selective MC4R agonist, displaced [¹²⁵I]-NDP- α -MSH (100 pM; $K_i = 199 \pm 37 \text{ nM}$, mean $\pm$ S.E.M., $n = 3$) from MC4Rs expressed in HEK-293 cells (Supplementary Fig. 2a). In contrast, clozapine only partially inhibited radioligand binding at the highest concentrations tested (10 and 50 μM), while olanzapine and risperidone, two other commonly prescribed APDs, had no effect (Supplementary Fig. 2a). These findings suggest that these APDs do not directly interact with the orthosteric binding site of MC4R.

The canonical MC4R signaling cascade involves G_{α_s} -mediated activation of adenylyl cyclase and subsequent elevation of intracellular cyclic adenosine monophosphate (cAMP). To determine whether APD exposure alters this pathway, we measured α -MSH-evoked cAMP responses in HEK-293 cells expressing MC4R and a genetically encoded cAMP sensor³⁶. Clozapine, risperidone, and olanzapine (1 μM) did not affect α -MSH potency, efficacy, or the slope of the α -MSH–cAMP concentration–response curves, indicating that these APDs do not perturb MC4R– G_{α_s} coupling or downstream adenylyl cyclase activation (Supplementary Fig. 2b–d).

Clozapine and risperidone enhance MC4R–Kir7.1 coupling

Silencing *Mc4r*-expressing neurons in the paraventricular nucleus of the hypothalamus (hereafter referred to as MC4R^{PVH} neurons) induces hyperphagia and obesity in mice^{37,38}. Interestingly, acute application of risperidone to murine hypothalamic slices inhibits these neurons in an MC4R-dependent manner¹⁶. To determine whether risperidone and clozapine exert similar effects in vivo, we injected adeno-associated viruses (AAV) expressing a Cre-dependent fluorescent calcium indicator (GCaMP8s) into the PVH of *Mc4r-Cre* mice (Fig. 3a). Fiber photometry recordings in freely moving mice revealed that an intraperitoneal (i.p.) dose of clozapine (3 mg/kg, Fig. 3b) or risperidone (3 mg/kg, Fig. 3c) suppressed MC4R^{PVH} neuron activity. In contrast, ziprasidone (3 mg/kg), an APD with minimal propensity to cause weight gain in patients³⁹, had no effect (Fig. 3d).

Previous work demonstrated that risperidone inhibits MC4R^{PVH} neurons by activating postsynaptic potassium conductance¹⁶. However, the specific potassium channel involved remained unknown. Notably, MC4R^{PVH} neurons express the inwardly rectifying potassium channel Kir7.1 (encoded by the *kcnj13* gene)^{40,41}. In HEK293 cells co-expressing MC4R and Kir7.1, AgRP increases Kir7.1 conductance in an MC4R-dependent manner^{40,42}. To test whether APDs modulate this pathway, we used a high-throughput automated patch-clamp platform (SyncroPatch 384PE; Nanion Technologies, Munich, Germany)⁴³ to measure MC4R–Kir7.1-dependent potassium currents in HEK-293 cells. Peak current ratios (I_{compound}/I_0) were used to quantify compound-induced channel activation across a range of APD concentrations. Both clozapine and risperidone increased inward potassium currents in a concentration-dependent manner (0.2 nM–20 μM), with pEC₅₀ values of 1.86 nM and 0.28 nM, respectively (Fig. 3e, f and

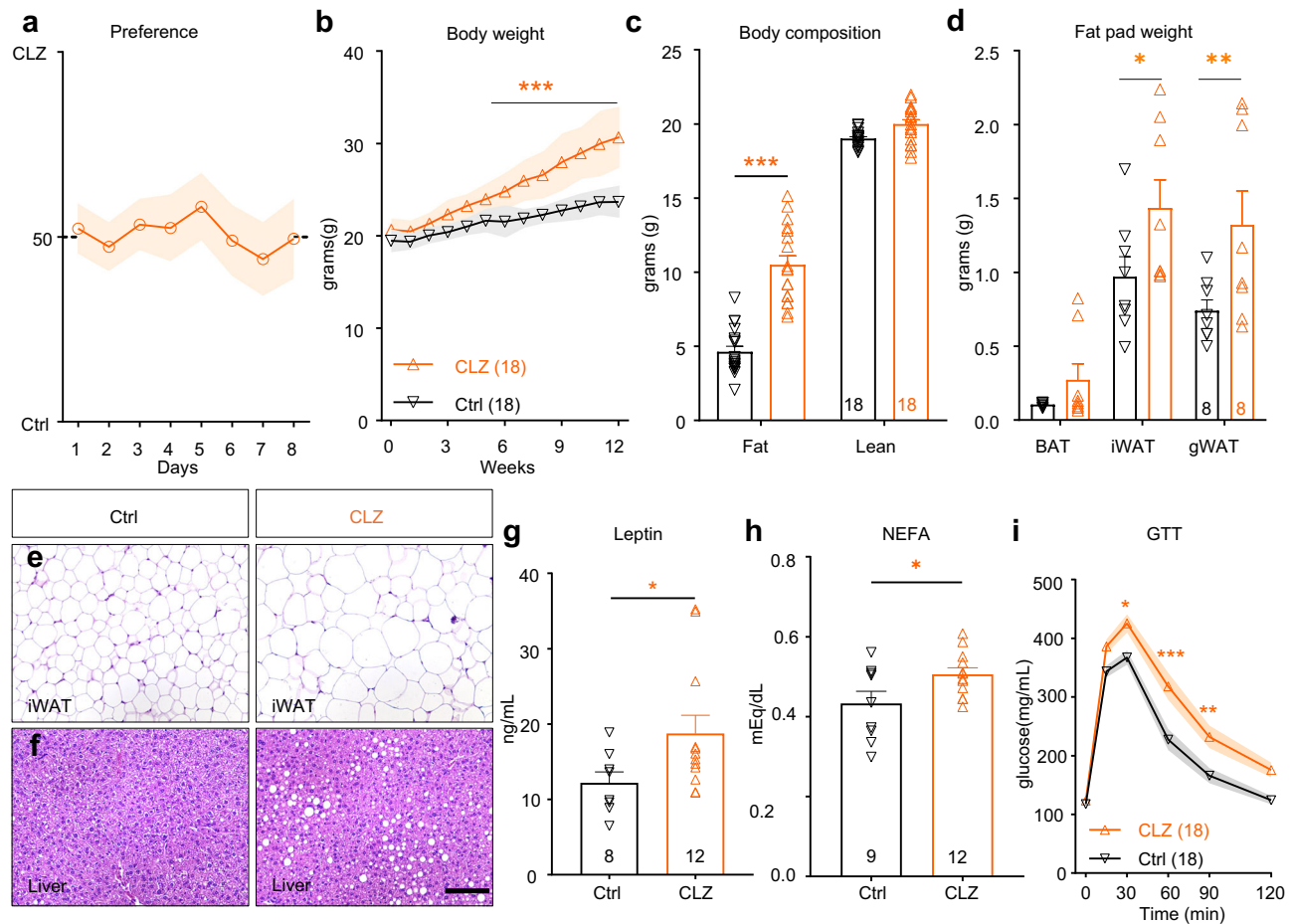


Fig. 1 | Clozapine treatment causes excessive weight gain in female C57BL/6 mice. **a** Preference ratio of daily intake of clozapine (CLZ) diet (50 mg/kg) relative to control (Ctrl) diet when both diets were available for 8 days ($n = 10$ mice). **b** Body weight of wild-type C57BL/6 mice fed the Ctrl or CLZ diet for 12 weeks ($n = 18$ mice; two-way ANOVA, $F(1,34) = 49.53$, $p = 4.0 \times 10^{-8}$). **c** Body composition after 12 weeks of Ctrl or CLZ diet treatment ($n = 18$ mice; two-way ANOVA, $F(1,68) = 39.7$, $p = 2.5 \times 10^{-8}$). **d** Fat pad weight in mice treated with the Ctrl or CLZ diet ($n = 8$ mice; two-way ANOVA, $F(2,42) = 11.77$, $p = 0.001$). BAT brown adipose tissue, iWAT

inguinal white adipose tissue, gWAT gonadal white adipose tissue.

e, f Representative H&E staining of adipose tissue from mice after 12 weeks on Ctrl or CLZ diet. Scale bar, 100 μ m. Plasma levels of leptin (**g**) and non-esterified fatty acids (NEFA) (**h**) after 12 weeks of Ctrl or CLZ diet treatment ($n = 8-12$ mice; Two-tailed unpaired t-test; $P = 0.0387$ for **g** and $P = 0.0374$ for **h**). **i** Glucose tolerance test (GTT) following 12 weeks of Ctrl or CLZ diet treatment ($n = 18$ mice; two-way ANOVA, $F(1,34) = 10.57$, $p = 0.003$). * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$; two-way ANOVA with Sidak's post hoc test. Data are presented as mean $\pm$ SEM.

Supplementary Fig. 3a, b). In contrast, olanzapine did not increase potassium conductance but instead triggered channel closure (decreased potassium currents compared to the reference) at the highest concentration tested (10 μ M; pIC_{50} of 1.95 μ M; Supplementary Fig. 4a).

Importantly, neither α -MSH, AgRP¹², nor any of the APDs elicited potassium currents in HEK-293 cells expressing Kir7.1 alone (Supplementary Fig. 5a–c), confirming that channel activation required the presence of MC4R. We next examined whether APDs modulate AgRP-induced Kir7.1 opening⁴⁴. Co-application of clozapine or risperidone (1 μ M each) enhanced AgRP-induced MC4R–Kir7.1 coupling by approximately 50%, without affecting AgRP potency, efficacy, or the slope of the concentration–response curves (Fig. 3g, h). In comparison, olanzapine (1 μ M) had no effect on AgRP-induced opening of Kir7.1 (Supplementary Fig. 4b).

Kir7.1 is required for clozapine-induced weight gain

We previously demonstrated that MC4R expression in PVH neurons is required for risperidone-induced weight gain, and that the selective melanocortin agonist setmelanotide reverses hyperphagia and obesity in risperidone-treated mice¹⁶. Consistent with this, chronic setmelanotide treatment reduced food intake and body weight in

clozapine-treated mice (Supplementary Fig. 6). To determine the role of Kir7.1 in APD-induced weight gain, we generated *Kir7.1^{fl/fl}; Mc4r-Cre^{+/+}* mice (hereafter referred to as *Kir7.1^{Mc4r KO}* mice), in which Kir7.1 was selectively deleted from *Mc4r-Cre*-expressing neurons. In situ hybridization (RNAscopeTM) confirmed reduced Kir7.1 mRNA expression in MC4R^{PVH} neurons (Fig. 4a and Supplementary Fig. 7). Consistently, fiber photometry recordings showed diminished calcium responses in these neurons following an intraperitoneal dose of clozapine or risperidone (Fig. 4b, c). Notably, clozapine-induced weight gain was diminished in *Kir7.1^{Mc4r KO}* mice (Fig. 4d). In contrast, these mice remained susceptible to olanzapine-induced increases in body weight and adiposity (Fig. 4d, e). These findings indicate that Kir7.1 in *Mc4r-Cre* neurons is necessary for the obesogenic effect of clozapine but not for that of olanzapine.

We next tested whether pharmacological blockade of Kir7.1 could mitigate clozapine-induced hyperphagia and weight gain using ML418, a small-molecular channel blocker with relative specificity for Kir7.1⁴⁵. To this end, C57BL/6 mice were maintained on the clozapine diet for 12 weeks to establish hyperphagia and obesity and then treated daily with either ML418 (5 mg/kg, i.p.) or vehicle while continuing the clozapine diet. ML418 treatment reduced food intake (Fig. 4f) and lowered body weight after 10 days (Fig. 4g). ML418-treated mice also showed

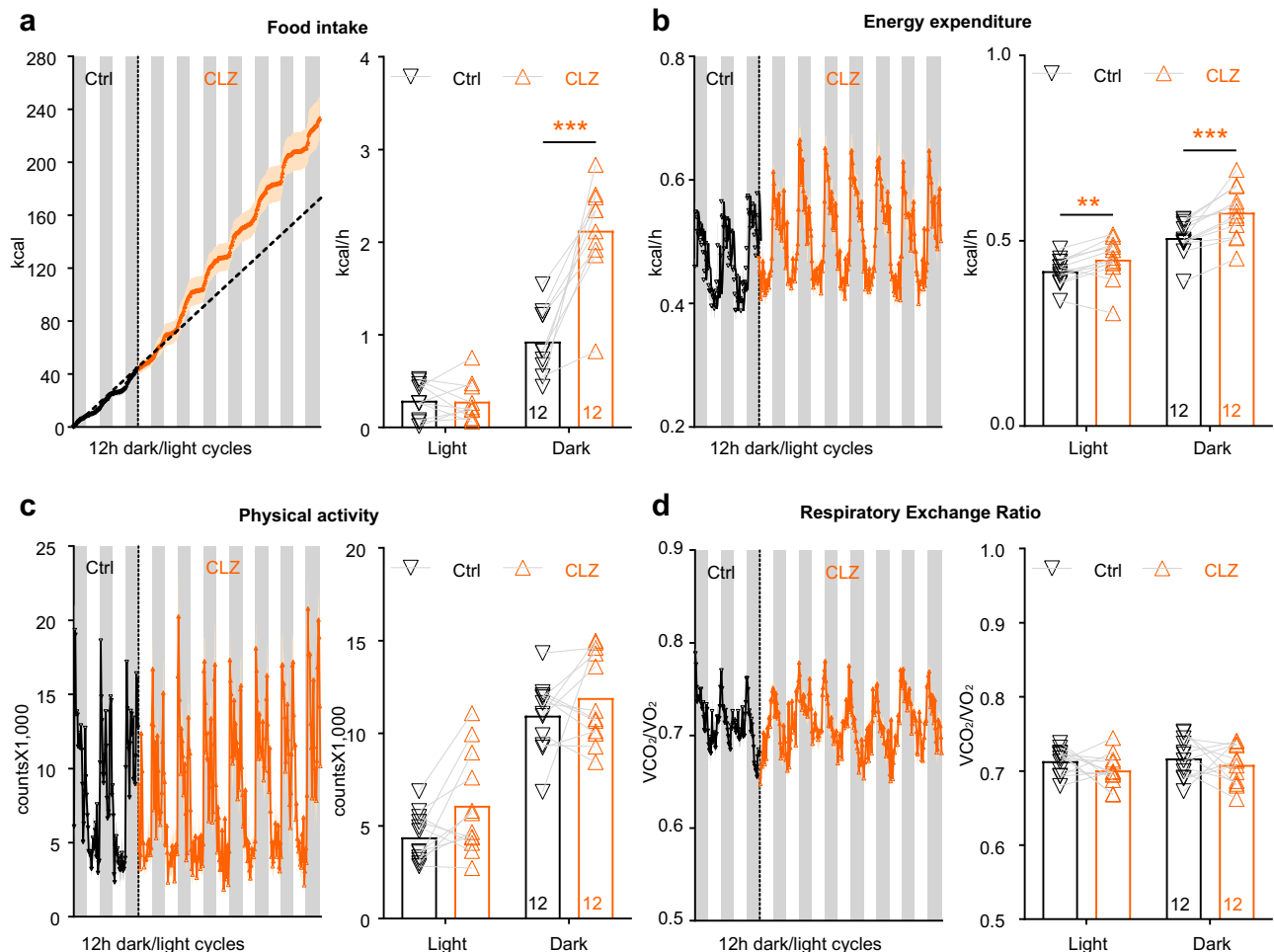


Fig. 2 | Clozapine-induced hyperphagia drives weight gain. **a–d** Left: representative traces of continuous measurements in metabolic cages; Right: summarized daily averages (binned into 12-hour light and dark phases) before (black) and after (orange) the dietary switch ($n = 12$ mice). The dashed line indicates the time point when the diet was switched to the clozapine diet. **a** Food intake

($F(1,9) = 30.83$, $p = 0.0004$). **b** Energy expenditure ($F(1,11) = 9.38$, $p = 0.01$, $p_{\text{light}} = 0.009$; $p_{\text{dark}} = 1.5 \times 10^{-5}$). **c** Physical activity ($F(1,11) = 0.35$, $p = 0.57$). **d** Respiratory exchange ratio (RER) ($F(1,11) = 0.20$, $p = 0.66$). ** $p < 0.01$, *** $p < 0.001$; two-way ANOVA with Sidak's post hoc test. Data are presented as mean $\pm$ SEM.

reduced fat mass (Fig. 4h), lower plasma leptin levels (Fig. 4i), and improved glucose tolerance (Fig. 4j).

Finally, we examined whether ML418 interferes with clozapine's antipsychotic efficacy using a rodent model of schizophrenia⁴⁶. Amphetamine is well known to induce psychosis-like symptoms in healthy human subjects⁴⁷ and to exacerbate symptoms in patients with schizophrenia⁴⁸. Similarly, administration of amphetamine (2.5 mg/kg, i.p.) induced robust hyperactivity in C57BL/6 mice, which was significantly attenuated by pretreatment with clozapine (Fig. 4k). Importantly, we found that co-treatment with ML418 did not diminish clozapine's ability to suppress amphetamine-induced hyperlocomotion in an open-field test (Fig. 4k). Together, these results demonstrate that Kir7.1 mediates clozapine-induced metabolic dysfunction without impairing its antipsychotic action, highlighting Kir7.1 as a promising therapeutic target to dissociate the metabolic and psychotropic effects of clozapine.

Discussion

Our study identifies a molecular mechanism underlying clozapine-induced weight gain and highlights MC4R-Kir7.1 coupling as a potential therapeutic target for mitigating antipsychotic-induced metabolic syndrome. Using a dietary administration paradigm, we established a mouse model that recapitulates key features of clozapine's metabolic effects, including hyperphagia, increased adiposity, and impaired

glucose homeostasis. This model provides a robust platform for dissecting the neural circuits and molecular pathways that drive clozapine-induced obesity.

We found that hyperphagia emerges shortly after clozapine exposure, indicating that excessive energy intake is the primary driver of weight gain. Clozapine also enhances energy expenditure, consistent with previous reports of increased sympathetic tone and thermogenesis in brown adipose tissue⁴⁹. However, the modest rise in energy expenditure is insufficient to offset the substantial increase in food intake, leading to a net positive energy balance and obesity.

Our findings implicate suppression of the central melanocortin system as a contributor to clozapine-induced hyperphagia. Specifically, we show that clozapine and risperidone inhibit the activity of MC4R^{PVH} neurons, a key neuronal population governing energy homeostasis. Moreover, the propensity of different APDs to induce weight gain may correlate with their ability to inhibit these neurons. Consistent with this idea, ziprasidone, an APD with minimal metabolic side effects, does not suppress MC4R^{PVH} neuronal activity. Although clozapine's polypharmacology almost certainly contributes to its overall metabolic impact, our data identify the central melanocortin system as a necessary downstream pathway through which clozapine drives hyperphagia.

Mechanistically, clozapine-induced inhibition occurs independently of canonical MC4R-G α s signaling and instead results from

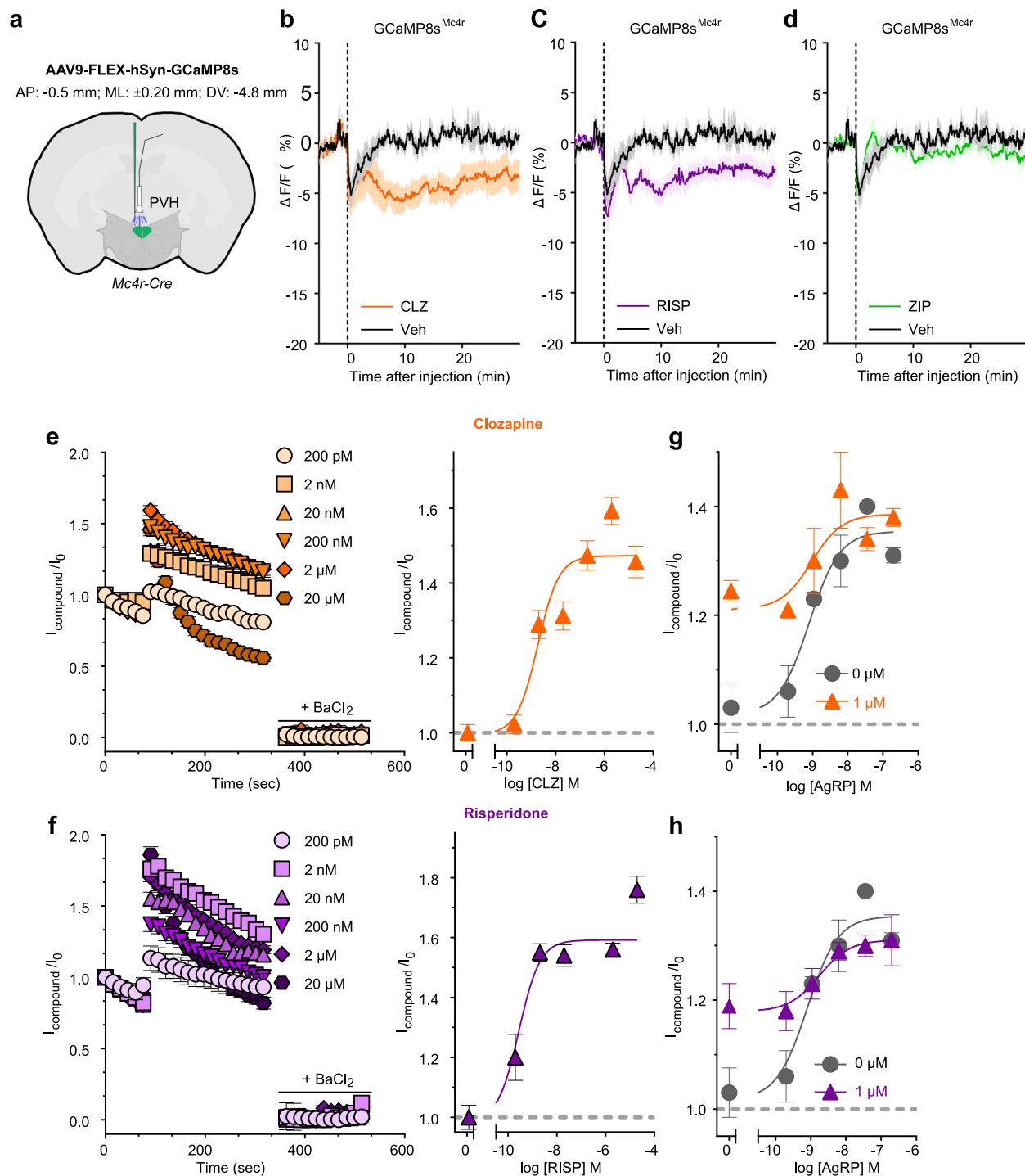


Fig. 3 | Clozapine and risperidone promote the Kir7.1 open state. **a** Schematic showing unilateral AAV9-FLEX-GCaMP8s injections into the paraventricular nucleus of the hypothalamus (PVH) of *Mc4r-Cre* mice. Calcium responses in MC4R^{PVH} neurons of living mice following intraperitoneal (i.p.) injection of clozapine (CLZ; **b**), risperidone (RIS; **c**), ziprasidone (ZIP; **d**), or vehicle (Veh) ($n = 7$). Dashed lines indicate the time of injection. **e, f** Left: representative whole-cell recordings of compound-evoked currents relative to baseline (I_{compound}/I_0) at increasing concentrations (100 pmol–10 μ mol). Right: concentration–response curves for clozapine (**e**) and risperidone (**f**) in TREx HEK-293 cells co-expressing MC4R and Kir7.1. Data points represent mean $\pm$ SEM from 5–29 wells per condition across three independent experiments. For **e** and **f**, n values are listed in order of

increasing drug concentration: 200 pM, 2 nM, 20 nM, 200 nM, 2 μ M, and 20 μ M. In **e**, $n = 15, 18, 20, 24, 16$, and 16 , respectively; in **f**, $n = 5, 19, 28, 27, 29$, and 10 , respectively. Agouti-related peptide (AgRP) concentration–response curves in the absence or presence of 1 μ M clozapine (**g**) or risperidone (**h**). Data points represent mean $\pm$ SEM from 7–13 wells per condition across two independent experiments. For **g** and **h**, n values are listed in order of increasing AgRP concentration. In **g**, $n = 13, 12, 12, 9, 12$, and 13 for the control group and $n = 7, 11, 7, 9, 10$, and 9 for the CLZ group. In **h**, $n = 9, 9, 9, 9, 12$, and 9 for the control group and $n = 11, 12, 12, 9, 10$, and 14 for the RIS group.

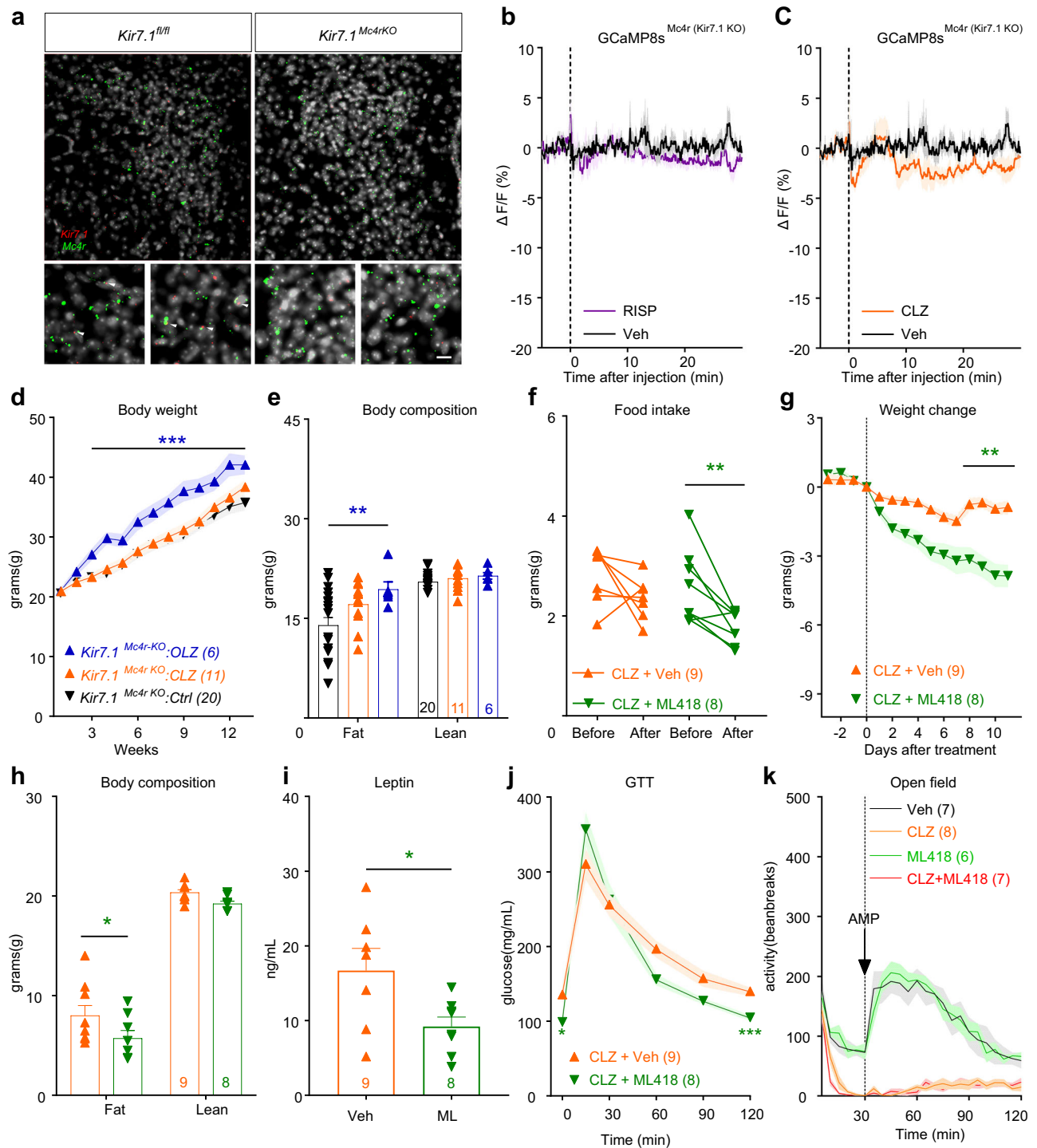


Fig. 4 | *Kir7.1* in *Mc4r-Cre* neurons is required for clozapine-induced weight gain. **a** RNAscope™ analyses showing *Mc4r* (green) and *Kir7.1* (red) mRNA expression in the paraventricular nucleus of the hypothalamus (PVH) of *Kir7.1^{fl/fl}* and *Kir7.1^{Mc4rKO}* mice. Scale bar, 10 μ m. Calcium responses in MC4R^{PVH} neurons of *Kir7.1^{Mc4rKO}* mice following intraperitoneal (i.p.) injection of risperidone (RISP; **b**), clozapine (CLZ; **c**), or vehicle (Veh) ($n = 5$). Dashed lines indicate the time of injection; two-way ANOVA, $F(2,9) = 1.232$, $p = 0.34$. **d** Body weight of *Kir7.1^{Mc4rKO}* mice fed the control (Ctrl), olanzapine (OLZ), or clozapine (CLZ) diet ($n = 6-20$; two-way ANOVA, $F(2,34) = 5.44$, $p = 0.0089$). **e** Body composition of *Kir7.1^{Mc4rKO}* mice after treatment with Ctrl, OLZ, or CLZ diet ($n = 6-20$; two-way ANOVA, $F(2,68) = 5.23$, $p = 0.0077$). **f** Change in daily food intake in clozapine-fed C57BL/6 mice treated with ML418 or vehicle (Veh) ($n = 8-9$ mice; Two-tailed paired t-test,

$p = 0.0092$, $**p < 0.01$). **g** Change in body weight before and after ML418 or Veh treatment ($n = 8-9$; two-way ANOVA, $F(1,15) = 18.37$, $p = 0.0006$). **h** Body composition in clozapine-fed mice after ML418 or Veh treatment ($n = 8-9$; two-way ANOVA, $F(1,15) = 4.78$, $p = 0.045$). **i** Plasma leptin levels in clozapine-fed mice after ML418 or Veh treatment ($n = 8-9$; Two-tailed unpaired t-test, $p = 0.0306$). **j** Glucose tolerance test (GTT) in clozapine-fed mice after ML418 or Veh treatment ($n = 8-9$; two-way ANOVA, $F(1,15) = 1.59$, $p = 0.23$, $p_0 = 0.047$, $p_{120} = 0.00096$). **k** Open-field test measuring amphetamine-induced hyperlocomotion in mice pretreated with Veh, CLZ, ML418, or CLZ + ML418 ($n = 6-8$; two-way ANOVA, $F(69,552) = 10.31$, $p < 0.0001$). $*p < 0.05$, $**p < 0.01$, $***p < 0.001$; two-way ANOVA with Sidak's post hoc test. Data are presented as mean $\pm$ SEM.

enhanced coupling between MC4R and the Kir7.1 potassium channel. Patch-clamp electrophysiology shows that clozapine and risperidone dose-dependently increase Kir7.1-mediated inward potassium currents, consistent with membrane hyperpolarization and reduced cellular excitability. These recordings were performed in heterologous HEK-293 cells, which lack the complex signaling environment of hypothalamic neurons. Although MC4R–Kir7.1 coupling has been established in native PVH neurons⁴⁰, whether APDs engage this coupling in the endogenous neuronal context remains to be determined.

Importantly, clozapine- and risperidone-induced neuronal inhibition is abolished in *Kir7.1^{Mc4r}KO* mice, which are resistant to clozapine-induced weight gain. These findings support a model in which certain APDs hijack melanocortin signaling by potentiating MC4R–Kir7.1 coupling to inhibit MC4R neurons, thereby contributing to hyperphagia and obesity. Our results also indicate that different APDs engage distinct neural mechanisms to induce weight gain. Although the obesogenic effect of olanzapine requires MC4R in PVH neurons¹⁶, it does not alter MC4R–Kir7.1 coupling. Consistent with this, olanzapine still causes significant weight gain in *Kir7.1^{Mc4r}KO* mice. Instead, olanzapine increases *Agrp* expression²⁶, which may secondarily inhibit MC4R^{PVH} neurons through enhanced AgRP-mediated antagonism⁴⁰.

Pharmacological inhibition of Kir7.1 with ML418 effectively reversed clozapine-induced hyperphagia and obesity, underscoring its therapeutic potential. Crucially, ML418 did not diminish clozapine's antipsychotic efficacy, as clozapine continued to suppress amphetamine-induced hyperlocomotion in ML418-treated mice. Together, these findings support a role for Kir7.1 in mediating clozapine-induced metabolic dysfunction and identify MC4R–Kir7.1 signaling as a potential target to mitigate APD-associated metabolic side effects without compromising psychiatric efficacy.

In a companion study, Peisley et al. define the structural basis for ML418's inhibition of Kir7.1. Using cryo-EM, they identify the ML418 binding site on Kir7.1 and demonstrate that intracerebroventricular administration of ML418, but not the selective Kir6.2 inhibitor VU0134992, activates MC4R^{PVH} neurons and suppresses food intake in wild-type mice, consistent with a major central role for Kir7.1. A recognized limitation of ML418 is its additional activity against Kir6.2/SUR1, the ATP-sensitive potassium (KATP) channel complex⁵⁰, and because ML418 was administered systemically (i.p.) in our study, peripheral contributions from Kir6.2/SUR1 inhibition cannot be formally excluded. However, the directionality of KATP channel pharmacology argues against a meaningful contribution of peripheral KATP inhibition to the phenotype observed here. Sulfonylureas, which inhibit pancreatic β -cell Kir6.2/SUR1, enhance insulin secretion and are clinically associated with weight gain^{51,52}. Conversely, the KATP channel activator diazoxide choline (Vykat XR) was recently approved by the FDA for treatment of hyperphagia in Prader-Willi syndrome, where it reduces food intake and fat mass⁵³. Peripheral Kir6.2/SUR1 inhibition by ML418 would therefore be expected to oppose, not mimic, its observed anorexigenic and weight-reducing effects. Together with the i.c.v. data from Peisley et al., this supports attribution of ML418's metabolic effects in our paradigm to a major role for central Kir7.1, consistent with our genetic findings in *Kir7.1^{Mc4r}KO* mice. Definitive attribution will benefit from future studies using more selective Kir7.1 pharmacology as they become available.

In conclusion, we identify MC4R–Kir7.1 coupling as a critical downstream pathway necessary for clozapine-induced weight gain. These findings provide mechanistic insight into APD-induced metabolic syndrome and highlight this signaling axis as a potential therapeutic target for preventing or reversing antipsychotic-associated obesity. However, our data should not be interpreted as supporting an exclusive role for the MC4R–Kir7.1 complex in APD-induced metabolic dysfunction. Clozapine exhibits broad polypharmacology, including high-affinity antagonism at histamine H1 and serotonin 5-HT_{2C} receptors⁵⁴, both of which have been implicated in antipsychotic-

induced weight gain^{14,15}. It is therefore likely that clozapine's obesogenic effects arise from the combined engagement of multiple parallel and intersecting neural pathways. Moreover, because Kir7.1 is expressed in peripheral tissues, systemic inhibition may raise safety concerns. Future studies should evaluate the translational feasibility of targeting Kir7.1, including the development of tissue-specific or centrally restricted modulators to minimize off-target effects, as well as strategies that selectively disrupt MC4R–Kir7.1 coupling. Determining whether this pathway contributes to metabolic disturbances induced by other APDs or across broader forms of obesity will further clarify its therapeutic potential.

Methods

Animals

All mice were housed in a temperature-controlled room in the UT Southwestern Medical Center animal facility with a 12-hour light/12-hour dark cycle (lights on at 06:00 am, lights off at 6:00 pm), with the temperature maintained at 68–79 °F (20–26 °C) and relative humidity maintained at 30–70%. Unless otherwise noted, mice were provided standard chow (No. 2016; Harlan Teklad) and water ad libitum. C57BL/6 mice were purchased from the Jackson Laboratory (000664). The *Kir7.1^{fl/fl}* mouse line⁵⁵ was generously provided by Dr. Graeme Mardon at Baylor College of Medicine. *Mc4r-Cre* +/-; *Kir7.1^{fl/fl}* and *Kir7.1^{fl/fl}* littermates were maintained on a C57BL/6 background. The Institutional Animal Care and Use Committee of the University of Texas Southwestern Medical Center approved all mouse experiments. For i.p. injection, clozapine (Sigma, Cat #C6305), risperidone (Acros Organics, Cat #AC459390010), olanzapine (Sigma, Cat #O1141), ziprasidone (Sigma, Cat #1724408), and ML418 (MedChemExpress, Cat #HY-122697) were first dissolved in DMSO to a final concentration of 10%. The drug solutions were then mixed with PEG400 to a final concentration of 20% before being combined with saline. Chronic delivery of setmelanotide was achieved using subcutaneous osmotic minipumps (Alzet). Pumps were implanted under isoflurane anesthesia, followed by postoperative analgesia with carprofen. Setmelanotide (2 mg/kg/day) was delivered continuously throughout the experimental period. All animal experiments were approved by the Institutional Animal Care and Use Committee at UT Southwestern Medical Center.

Genotyping

To genotype *Mc4r-Cre*^{+/+}; *Kir7.1^{fl/fl}* and *Kir7.1^{fl/fl}* mice, the following primers were used: *Mc4r-Cre* primers, F/s1 (GGCAAACGGACAGAAGCA), R/s1 (TCCGGAGTCAAGAAGTGAAG), and R/s2 (CAATTCATAACGCCACACT); and *Kir7.1^{fl/fl}* primers, F/s2 (AAGGGGACATGGTGGCTTG), R/s3 (ACCATTGGCAGGGAAGTCAA), F/s3 (CAATGAAACCACTCAACAACC), and R/s4 (CTGGAAATAGCACAGTCTCC).

Diets

All diets were prepared by ResearchDiets Inc. and shared the same macronutrients, ingredient composition, and energy density (4.76 kcal/gram). Clozapine (MedChemExpress, Cat# HY-14539) was mixed into a control diet D09092903 to make the clozapine diet D21070104 (50 mg/kg). Similarly, olanzapine (ChemPacif, Cat# 356941G) was mixed into the same control diet, to make the olanzapine diet D12040807 (50 mg/kg)¹⁴.

Metabolic phenotype analysis

We used a magnetic-resonance whole-body composition analyzer (EchoMRI) to analyze the mouse's body composition (fat mass, lean mass, and water content). For chronic studies, body weight and food intake were measured weekly. The acute effects of clozapine on energy intake and expenditure were measured using an indirect calorimetric system (TSE PhenoMaster, Germany) in the Metabolic Phenotyping Core of UT Southwestern Medical Center.

Radioligand binding experiments

Radioligand binding experiments used a HEK-293 (ATCC CRL-1573™) stable cell line expressing the human MC4R. The cells were grown in 150 mm culture dishes to prepare crude membranes until they reached 90–95% confluency. TrypLE Express (1x) cell dissociation reagent (Thermo Fisher Scientific) was used to detach the cells from the growth vessels. Afterward, trypsin was neutralized with an equal volume of complete growth media. The cells were pelleted at $200 \times g$ for 5 min, washed twice with phosphate-buffered saline (PBS), and resuspended in “Membrane Extract Buffer” (ME-buffer). This buffer contained 50 mM Tris-HCl (pH 7.4), 10 mM MgSO₄, 1.5 mM CaCl₂, 0.5 mM EDTA, and cOmplete™ EDTA-free protease inhibitor cocktail (Roche, Basel Switzerland). The initial cell pellet was homogenized using a 15 mL Dounce homogenizer. Homogenates were centrifuged at $35,000 \times g$ for 20 min, and the process was repeated twice. The final pellet was resuspended with $10 \times v/w$ of ME-buffer supplemented with 10% sucrose, and aliquots were stored at -80°C . The total protein content from the crude membrane extracts was determined using the bicinchoninic acid assay⁵⁶ with a commercially available kit (Pierce™ BCA Protein Assay Kit, Thermo Fisher Scientific).

Equilibrium competition binding experiments were performed with 4 μg of total protein from the crude membrane extract. [¹²⁵I]-[Nle⁴, DPhe⁷]- α -MSH (NDP-MSH, specific activity: 2200 Ci/mmol) was used as the radioligand tracer at a concentration of 100 pM. This radioligand concentration resembles the equilibrium dissociation constant (K_D) determined by ligand saturation experiments. Varying concentrations of the indicated antipsychotic drugs (APD) or PF-00446687, an MC4R-specific agonist, were incubated with the crude membrane extracts in a final volume of 200 μL of Binding Assay buffer (BA-buffer: 25 mM HEPES pH 7.4, 1 mM MgSO₄, 1 mM CaCl₂, 0.2% bovine serum albumin (BSA), and cOmplete™ EDTA-free protease inhibitor cocktail). APD and crude membranes were preincubated for 16 hours at room temperature before adding radioligand to ensure equilibrium. The assay was incubated for an additional 120 min in the presence of the radioligand. Bound [¹²⁵I]-NDP-MSH was rapidly separated from the free ligand by filtration through GF/B glass fiber filters (Brandel, Gaithersburg, MD) using a cell harvester (Brandel) manifold. During filtration, the filters were washed five times rapidly with cold 25 mM Tris-HCl pH 7.4. IC₅₀ values were derived from nonlinear regression fitting the data to a sigmoid function model. Ki values were derived by applying the Cheng and Prusoff transformation⁵⁷.

Cellular cAMP responses

A HEK-293 (ATCC CRL-1573™) stable cell line expressing the human MC4R and a split luciferase cAMP genetic sensor based on the circularized fusion of firefly luciferase and protein kinase A regulatory cAMP binding domain (clone 22F, Promega, Madison WI) was used to determine the α -MSH-elicited cAMP responses in live cells as previously described⁴⁴. The cell line identity was verified by qPCR and MC4R subtype-specific oligonucleotides⁵⁸.

Briefly, twenty-four hours before the cAMP response recordings, the cells were plated in poly-D-lysine coated clear-bottom black wall 384-well plates (Corning Inc. Corning, NJ) at a density of 20,000 cells per well in growth medium (Dulbecco's Modified Eagle Medium (DMEM) supplemented with 4.5 g/L D-glucose, 110 mg/L sodium pyruvate, 10% fetal bovine serum, 1 $\times$ Antibiotic-Antimycotic™ cocktail (Thermo Fisher Scientific, Waltham, MA), 200 $\mu\text{g}/\text{mL}$ hygromycin B, and 700 $\mu\text{g}/\text{mL}$ Geneticin™). Two hours before the start of the experiment, the growth medium was replaced with CO₂ Independent Medium™ (Thermo Fisher Scientific), containing 4% D-luciferin (Promega). The luciferase substrate permeated the cells for 120 min at 37°C . Intracellular cAMP levels were measured using an FDSS 7000EX Functional Drug Screening System (Hamamatsu Photonics, Hamamatsu, Japan). To test the effect of APD on the α -MSH concentration-

response profile, the in-line addition of 1 μM (final concentration) clozapine, risperidone, or olanzapine was performed and allowed to reach equilibrium for 12 minutes. This was followed by adding 10 μL of varying $3 \times$ concentrations of α -MSH or vehicle. Luminescence responses were measured for an additional 12 min. The agonist responses were normalized to the α -MSH maximum response without APD. The data were fit to a four-parameter sigmoid model, constraining the slope to a range of 0 to 2 using GraphPad Prism.

Histology

Details for performing these experiments were described previously^{16,59}. For brain collection, mice were deeply anesthetized with chloral hydrate (200 mg/kg, i.p.) and euthanized by transcardial perfusion with phosphate-buffered saline followed by 4% paraformaldehyde. Brains were removed, post-fixed overnight in 4% paraformaldehyde, and sectioned using a vibratome (Leica VT1000S). RNAscope was performed using ACD RNAscope Multiplex Fluorescent Detection Kit version 2 following the manufacturer's (Advanced Cell Diagnostics) protocol with the following probes Mm-Mc4r and Mm-Kir7.1.

Mouse adipose tissue and liver were fixed in 4% paraformaldehyde overnight and then dehydrated through a series of ethanol baths with ascending concentrations (up to 100%). The Histo Pathology Core of the University of Texas Southwestern Medical Center performed paraffin embedding, sectioning, and H&E staining.

Automated patch clamp

Patch clamp recordings were conducted using a SyncroPatch 384PE system from Nanion Technologies (Munich, Germany) at the Center for Chemical Genomics at the University of Michigan. Recordings were carried out in the whole cell voltage clamp configuration employing medium resistance four-hole NPC-384 chips. Data acquisition and analysis were performed with PatchControl 384 (for data acquisition) and DataControl 384 (for data analysis), both from Nanion Technologies. The experimental setup utilized an inducible TREx HEK-293 cell line (Thermo Fisher Scientific) expressing MC4R and the inwardly rectifying K⁺ channel Kir7.1 mutant M125R. Cells were centrifuged at $200 g$ for 5 minutes, resuspended in an external solution at a density of 500,000 cells/mL, and added to the instrument cell hotel. The cell hotel was maintained at 10°C with 200 rpm agitation until the start of the experiment. The external recording solution comprised (in mM): NaCl 140, KCl 4, CaCl₂ 2, MgCl₂ 1, HEPES 10, glucose 5; pH 7.4 (NaOH, ~ 300 mOsm). Cell sealing was accomplished in a modified external solution containing 6 mM CaCl₂. The internal electrode solution included (in mM): KF 110, KCl 10, NaCl 10, HEPES 10, EGTA 10; pH 7.2 (KOH, ~ 280 mOsm).

To assess MC4R-dependent Kir7.1 channel closures or activations induced by antipsychotic drugs (APD), a two-stage voltage protocol was applied. This protocol involved a hyperpolarization step to -140 mV for 200 ms, starting from a holding potential of -60 mV, and it was repeated every 5 seconds. Peak Kir7.1 inward current amplitude during the hyperpolarizing step was measured and normalized to the remnant currents after adding 10 mM BaCl₂ (full block). Antipsychotic drugs (APD) were delivered at various concentrations across the NPC-384 chip, followed by a Ba²⁺ full channel block added to each well to determine non-K⁺ channel currents, as the M125R Kir7.1 regains sensitivity to Ba²⁺ compared to the relatively insensitive wild-type Kir7.1. Responses were further normalized by determining the ratios of peak currents in the presence or absence of APDs (I_{compound}/I_0). Concentration-response profiles were determined, and EC₅₀ or IC₅₀ values for activating or inhibiting APD responses were calculated. A seal quality assessment for each well was performed using the DataControl 384 application, excluding rundown wells, with a positive patch rate of 87%.

GTT

For the glucose tolerance test (GTT), mice were fasted for 7 hours with water provided *ad libitum* from 8 am on the experimental day. During GTT, blood glucose levels were monitored at 0, 15, 30, 60, 90, and 120 minutes after an i.p. dose of glucose (Dextrose) (2 g/kg body weight). Blood glucose was taken from the tail vein and analyzed using a Glucometer (Johnson & Johnson).

Stereotaxic surgery

Eight-week-old female mice were anesthetized with isoflurane. The skull was exposed and leveled on a stereotaxic frame (David Kopf Instruments). A small hole was drilled in the skull for viral injection or fiber implantation. AAV9-AAV-syn-FLEX-jGCaMP8s-WPRE (Addgene #162377-AAV9) was bilaterally injected into the PVH [anteroposterior (AP), −0.50 mm; mediolateral (ML), ±0.20 mm; dorsoventral (DV), −4.80 mm]. A total of 180 nL of the virus was injected at a rate of 30 nL/min and allowed to diffuse for eight to ten minutes before the injection needle was removed. After a three-week recovery period, a 1.25 mm ceramic ferrule-capped fiber (RWD, Cat# R-FOC-BL200C-39NA, 200 μm core, 0.39 NA) was unilaterally implanted for fiber photometry [anteroposterior (AP), −0.50 mm; mediolateral (ML), +0.10 mm; dorsoventral (DV), −4.80 mm]. As previously described⁶⁰, the injection site was verified by the jGCaMP8s fluorescent signal for each mouse.

Amphetamine-induced hyperlocomotion

The Rodent Behavior Core performed the amphetamine-induced hyperlocomotion in the Peter O'Donnell Jr. Brain Institute at UT Southwestern Medical Center. Briefly, C57BL/6 mice were habituated for the open field test with daily single i.p. saline injections for three days. On the third day of habituation, all mice were given D-amphetamine (2.5 mg/kg) 30 min after the saline injection. Following the habituation, all mice were randomly divided into four groups: 1) mice in the saline group received an i.p. dose of saline; 2) Mice in the ML418 group received an i.p. dose of ML418 (5 mg/kg); 3) mice in the CLZ group received an i.p. dose of clozapine (3 mg/kg); 4) mice in the ML418 + CLZ group received both clozapine (3 mg/kg) and ML418 (5 mg/kg). Locomotor activity (beam breaks) was immediately recorded after the first injection. An i.p. dose of D-amphetamine (2.5 mg/kg) was administered 30 min after the first injection.

Quantification and statistical analyses

Sample sizes were chosen to reliably measure experimental parameters while remaining in compliance with ethical guidelines for minimizing animal use, and they were similar to those reported in previous publications. All the virus expression and optic fiber implant placements were verified by *post hoc* histology. Replicate information is indicated in the figure legends. All results are presented as mean ± SEM and analyzed using statistical tools implemented in Prism (GraphPad, version 10). Statistical analyses were performed using the Student's *t*-test and regular one-way or two-way analysis of variance (ANOVA). Differences with *P* < 0.05 were considered to be significant. *P* < 0.05 (*), *P* < 0.01 (**), and *P* < 0.001 (***).

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The data supporting the findings of this study are available in the Source Data file. Source data are provided with this paper.

References

- Meltzer, H. Y. Update on typical and atypical antipsychotic drugs. *Annu. Rev. Med.* **64**, 393–406 (2013).

- Gohlke, J. M. et al. Recent advances in understanding and mitigating adipogenic and metabolic effects of antipsychotic drugs. *Front. Psychiatry* **3**, 62 (2012).
- Baptista, T. et al. Rosiglitazone in the assistance of metabolic control during olanzapine administration in schizophrenia: a pilot double-blind, placebo-controlled, 12-week trial. *Pharmacopsychiatry* **42**, 14–19 (2009).
- Baptista, T. et al. Metformin for prevention of weight gain and insulin resistance with olanzapine: a double-blind placebo-controlled trial. *Can. J. Psychiatry* **51**, 192–196 (2006).
- Cavazzoni, P., Tanaka, Y., Roychowdhury, S. M., Breier, A. & Allison, D. B. Nizatidine for prevention of weight gain with olanzapine: a double-blind placebo-controlled trial. *Eur. Neuropsychopharmacol.* **13**, 81–85 (2003).
- Kaar, S. J., Natesan, S., McCutcheon, R. & Howes, O. D. Anti-psychotics: mechanisms underlying clinical response and side-effects and novel treatment approaches based on pathophysiology. *Neuropharmacology* **172**, 107704 (2020).
- McCutcheon, R. A. et al. Data-driven taxonomy for antipsychotic medication: a new classification system. *Biol. Psychiatry* **94**, 561–568 (2023).
- Lange-Asschenfeldt, C., Brouzou, K. & Christl, J. Comparative receptor pharmacology of antipsychotic drugs based on normalized binding affinity data and breadth of interaction. *Eur. Neuropsychopharmacol.* **99**, 31–40 (2025).
- Meltzer, H. Y. & Huang, M. In vivo actions of atypical antipsychotic drug on serotonergic and dopaminergic systems. *Prog. Brain Res.* **172**, 177–197 (2008).
- Kim, D. H., Maneen, M. J. & Stahl, S. M. Building a better antipsychotic: receptor targets for the treatment of multiple symptom dimensions of schizophrenia. *Neurotherapeutics* **6**, 78–85 (2009).
- Olten, B. & Bloch, M. H. Meta regression: relationship between antipsychotic receptor binding profiles and side-effects. *Prog. Neuropsychopharmacol. Biol. Psychiatry* **84**, 272–281 (2018).
- Reynolds, G. P. & Kirk, S. L. Metabolic side effects of antipsychotic drug treatment-pharmacological mechanisms. *Pharm. Ther.* **125**, 169–179 (2010).
- Siafis, S., Tzachanis, D., Samara, M. & Papazisis, G. Antipsychotic drugs: from receptor-binding profiles to metabolic side effects. *Curr. Neuropharmacol.* **16**, 1210–1223 (2018).
- Lord, C. C. et al. The atypical antipsychotic olanzapine causes weight gain by targeting serotonin receptor 2C. *J. Clin. Invest.* **127**, 3402–3406 (2017).
- Kim, S. F., Huang, A. S., Snowman, A. M., Teuscher, C. & Snyder, S. H. From the cover: antipsychotic drug-induced weight gain mediated by histamine H1 receptor-linked activation of hypothalamic AMP-kinase. *Proc. Natl. Acad. Sci. USA* **104**, 3456–3459 (2007).
- Li, L. et al. The atypical antipsychotic risperidone targets hypothalamic melanocortin 4 receptors to cause weight gain. *J. Exp. Med.* **218** <https://doi.org/10.1084/jem.20202484> (2021).
- Su, Y. et al. Olanzapine modulate lipid metabolism and adipose tissue accumulation via hepatic muscarinic M3 receptor-mediated Alk-related signaling. *Biomedicines* **12** <https://doi.org/10.3390/biomedicines12071403> (2024).
- Ferno, J. et al. Olanzapine-induced hyperphagia and weight gain associate with orexigenic hypothalamic neuropeptide signaling without concomitant AMPK phosphorylation. *PLoS ONE* **6**, e20571 (2011).
- Wan, X. Q. et al. Risperidone stimulates food intake and induces body weight gain via the hypothalamic arcuate nucleus 5-HT2c receptor-NPY pathway. *CNS Neurosci. Ther.* **26**, 558–566 (2020).
- Pozzi, M. et al. Weight and body mass index increase in children and adolescents exposed to antipsychotic drugs in non-interventional settings: a meta-analysis and meta-regression. *Eur. Child Adolesc. Psychiatry* <https://doi.org/10.1007/s00787-020-01582-9> (2020).

21. Kaur, G. & Kulkarni, S. K. Studies on modulation of feeding behavior by atypical antipsychotics in female mice. *Prog. Neuropsychopharmacol. Biol. Psychiatry* **26**, 277–285 (2002).
22. Rimessi, A. et al. Protein kinase C beta: a new target therapy to prevent the long-term atypical antipsychotic-induced weight gain. *Neuropsychopharmacology* **42**, 1491–1501 (2017).
23. Nair, A. B. & Jacob, S. A simple practice guide for dose conversion between animals and human. *J. Basic Clin. Pharm.* **7**, 27–31 (2016).
24. Karampas, A. et al. Clozapine/norclozapine plasma concentrations and their ratio in treatment resistant, early psychosis patients. *Psychiatriki* **36**, 42–54 (2025).
25. Zhao, S. et al. Hyperleptinemia contributes to antipsychotic drug-associated obesity and metabolic disorders. *Sci. Transl. Med.* **15**, eade8460 (2023).
26. Zapata, R. C. et al. Nuclear receptor 5A2 regulation of Agrp underlies olanzapine-induced hyperphagia. *Mol. Psychiatry* <https://doi.org/10.1038/s41380-023-01981-9> (2023).
27. Chowdhury, N. I. et al. Genetic association study between antipsychotic-induced weight gain and the melanocortin-4 receptor gene. *Pharmacogenomics J.* **13**, 272–279 (2013).
28. Ellingrod, V. L. et al. Weight gain associated with the -759C/T polymorphism of the 5HT2C receptor and olanzapine. *Am. J. Med. Genet. B Neuropsychiatr. Genet.* **134B**, 76–78 (2005).
29. Park, Y. M. et al. Weight gain associated with the alpha2a-adrenergic receptor -1,291 C/G polymorphism and olanzapine treatment. *Am. J. Med. Genet. B Neuropsychiatr. Genet.* **141B**, 394–397 (2006).
30. Malhotra, A. K. et al. Association between common variants near the melanocortin 4 receptor gene and severe antipsychotic drug-induced weight gain. *Arch. Gen. Psychiatry* **69**, 904–912 (2012).
31. Zhang, J. P. et al. Pharmacogenetic associations of antipsychotic drug-related weight gain: a systematic review and meta-analysis. *Schizophr. Bull.* **42**, 1418–1437 (2016).
32. Maciukiewicz, M. et al. Genome-wide association study on antipsychotic-induced weight gain in Europeans and African-Americans. *Schizophr. Res.* **212**, 204–212 (2019).
33. Reynolds, G. P., Hill, M. J. & Kirk, S. L. The 5-HT2C receptor and antipsychotic-induced weight gain - mechanisms and genetics. *J. Psychopharmacol.* **20**, 15–18 (2006).
34. Sweeney, P., Gimenez, L. E., Hernandez, C. C. & Cone, R. D. Targeting the central melanocortin system for the treatment of metabolic disorders. *Nat. Rev. Endocrinol.* **19**, 507–519 (2023).
35. Lansdell, M. I. et al. Discovery of a selective small-molecule melanocortin-4 receptor agonist with efficacy in a pilot study of sexual dysfunction in humans. *J. Med. Chem.* **53**, 3183–3197 (2010).
36. Binkowski, B. F. et al. A luminescent biosensor with increased dynamic range for intracellular cAMP. *ACS Chem. Biol.* **6**, 1193–1197 (2011).
37. Garfield, A. S. et al. A neural basis for melanocortin-4 receptor-regulated appetite. *Nat. Neurosci.* **18**, 863–871 (2015).
38. Li, M. M. et al. The paraventricular hypothalamus regulates satiety and prevents obesity via two genetically distinct circuits. *Neuron* **102**, 653–667 e656 (2019).
39. Pillinger, T. et al. Comparative effects of 18 antipsychotics on metabolic function in patients with schizophrenia, predictors of metabolic dysregulation, and association with psychopathology: a systematic review and network meta-analysis. *Lancet Psychiatry* **7**, 64–77 (2020).
40. Ghamari-Langroudi, M. et al. G-protein-independent coupling of MC4R to Kir7.1 in hypothalamic neurons. *Nature* **520**, 94–98 (2015).
41. Anderson, E. J. P. et al. Late onset obesity in mice with targeted deletion of potassium inward rectifier Kir7.1 from cells expressing the melanocortin-4 receptor. *J. Neuroendocrinol.* **31**, e12670 (2019).
42. Yu, J. et al. Determination of the melanocortin-4 receptor structure identifies Ca(2+) as a cofactor for ligand binding. *Science* **368**, 428–433 (2020).
43. Obergrussberger, A. et al. Automated patch clamp meets high-throughput screening: 384 cells recorded in parallel on a planar patch clamp module. *J. Lab Autom.* **21**, 779–793 (2016).
44. Gimenez, L. E. et al. Novel cocrystal structures of peptide antagonists bound to the human melanocortin receptor 4 unveil unexplored grounds for structure-based drug design. *J. Med. Chem.* **67**, 2690–2711 (2024).
45. Swale, D. R. et al. ML418: the first selective, sub-micromolar pore blocker of Kir7.1 potassium channels. *ACS Chem. Neurosci.* **7**, 1013–1023 (2016).
46. Van Swearingen, A. E., Walker, Q. D. & Kuhn, C. M. Sex differences in novelty- and psychostimulant-induced behaviors of C57BL/6 mice. *Psychopharmacology* **225**, 707–718 (2013).
47. Janowsky, D. S. & Risch, C. Amphetamine psychosis and psychotic symptoms. *Psychopharmacology* **65**, 73–77 (1979).
48. Angrist, B., Rotrosen, J. & Gershon, S. Responses to apomorphine, amphetamine, and neuroleptics in schizophrenic subjects. *Psychopharmacology* **67**, 31–38 (1980).
49. Kalkman, H. O. & Loetscher, E. alpha2C-Adrenoceptor blockade by clozapine and other antipsychotic drugs. *Eur. J. Pharm.* **462**, 33–40 (2003).
50. Haider, S., Antcliff, J. F., Proks, P., Sansom, M. S. & Ashcroft, F. M. Focus on Kir6.2: a key component of the ATP-sensitive potassium channel. *J. Mol. Cell Cardiol.* **38**, 927–936 (2005).
51. Apovian, C. M., Okemah, J. & O'Neil, P. M. Body weight considerations in the management of type 2 diabetes. *Adv. Ther.* **36**, 44–58 (2019).
52. Nichols, C. G. KATP channels as molecular sensors of cellular metabolism. *Nature* **440**, 470–476 (2006).
53. Miller, J. L. et al. Diazoxide choline extended-release tablet in people with Prader-Willi syndrome: a double-blind, placebo-controlled trial. *J. Clin. Endocrinol. Metab.* **108**, 1676–1685 (2023).
54. Abela, A. R., Ji, X. D., Li, Z., Le, A. D. & Fletcher, P. J. Clozapine reliably increases the motivation for food: parsing the role of the 5-HT(2c) and H(1) receptors. *Psychopharmacology* **237**, 957–966 (2020).
55. Roman, D., Zhong, H., Yaklichkin, S., Chen, R. & Mardon, G. Conditional loss of Kcnj13 in the retinal pigment epithelium causes photoreceptor degeneration. *Exp. Eye Res.* **176**, 219–226 (2018).
56. Smith, P. K. et al. Measurement of protein using bicinchoninic acid. *Anal. Biochem.* **150**, 76–85 (1985).
57. Cheng, Y. & Prusoff, W. H. Relationship between the inhibition constant (K1) and the concentration of inhibitor which causes 50 per cent inhibition (I50) of an enzymatic reaction. *Biochem. Pharmacol.* **22**, 3099–3108 (1973).
58. Gimenez, L. E. et al. Demonstration of a common DPhe(7) to DNaI(2') (7) peptide ligand antagonist switch for melanocortin-3 and melanocortin-4 receptors identifies the systematic mischaracterization of the pharmacological properties of melanocortin peptides. *J. Med. Chem.* **65**, 5990–6000 (2022).
59. Xu, B. et al. Orthopedia regulates melanocortin 4 receptor transcription and energy homeostasis. *Sci. Transl. Med.* **17**, eadr6459 (2025).
60. Li, L. et al. Delineating a serotonin 1B receptor circuit for appetite suppression in mice. *J. Exp. Med.* **219** <https://doi.org/10.1084/jem.20212307> (2022).

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Author contributions

L.L., C.C.H., L.E.G., R.D.C., and C.L. designed experiments. L.L., C.C.H., L.E.G., B.X., N.S.D., A.P., W.Z., M.C., A.S., R.W., S., and S.G.B. collected data. L.L., C.C.H., L.E.G., R.D.C., and C.L. analyzed the data and wrote the manuscript.

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Competing interests

R.D.C., L.E.G., and the University of Michigan have equity in Courage Therapeutics, and R.D.C. serves on the company board. The other authors declare no competing interests.

Additional information

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